

Assignment 04: Ethics in Artificial Intelligence

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Section A: Conceptual Understanding

1. What is Ethical AI, and why is it essential in modern technological systems?

Ethical AI means creating and using AI systems in ways that respect people's rights, privacy, safety, and dignity. It also involves trying to prevent unfair outcomes and making sure humans stay responsible for important decisions. This matters because AI now affects things like hiring, healthcare, banking, education, and public safety. If a system is poorly designed, it can harm a lot of people quickly, while ethical systems help build trust and reduce unnecessary risks.

2. List and explain any four major ethical principles that guide AI development.

Four major principles are fairness, transparency, privacy, and accountability. Fairness means the system shouldn't discriminate based on things like race, gender, age, or disability. Transparency means being open about when AI is being used and what it's meant to do. Privacy focuses on handling personal data responsibly. Accountability means a person or organization must take responsibility for the system's decisions instead of blaming the algorithm.

3. What is the main objective of the IEEE Ethically Aligned Design framework?

The main goal of the IEEE Ethically Aligned Design framework is to put human well-being, rights, and values at the center of AI and autonomous system design. Instead of treating ethics as an afterthought, it encourages developers to consider ethical risks from the very beginning. The idea is that technology should serve people, not the other way around.

4. Define the term algorithmic bias and provide one real-world example.

Algorithmic bias happens when an AI system produces unfair or unequal results for certain groups. This can come from biased training data, design choices, mislabeled data, or variables that indirectly represent protected characteristics. A real example is facial-recognition technology. NIST found that some systems had higher error rates for certain demographic

groups, showing that even “neutral” systems can behave differently depending on who is being evaluated.

5. Explain the difference between transparency and explainability in AI.

Transparency is the broader idea of being open about how an AI system works — including what data it uses, who built it, and what its limitations are. Explainability focuses on a specific outcome and answers the question, “Why did the system make this decision?” For example, a bank can be transparent by saying it uses AI for loan approvals, but it’s only explainable if it can tell someone why their loan was denied.

6. What role does accountability play in the ethical use of AI systems?

Accountability ensures that humans and organizations remain responsible for AI outcomes. Without it, companies could avoid responsibility by blaming the algorithm. Ethical accountability requires clear ownership, documentation, testing, monitoring, and a way to fix harmful decisions. This is especially important in areas like healthcare, law enforcement, employment, and finance.

7. Identify three international organizations involved in developing AI ethics guidelines.

Three major organizations working on AI ethics are UNESCO, the OECD, and IEEE. UNESCO created a global recommendation focused on human rights, fairness, and sustainability. The OECD developed international principles for trustworthy AI. IEEE provides technical and ethical guidance to help engineers design systems that reflect human values.

8. What does UNESCO’s Recommendation on the Ethics of Artificial Intelligence emphasize?

UNESCO’s recommendation emphasizes protecting human rights, dignity, fairness, privacy, transparency, accountability, and human oversight. It also recognizes that AI affects more than just technology — it influences education, culture, healthcare, employment, the environment, and democracy. UNESCO encourages governments to create policies that allow society to benefit from AI while reducing risks like discrimination and surveillance abuse.

9. What are OECD's five key principles for trustworthy AI?

The OECD's five principles are inclusive growth and well-being; respect for human rights, democratic values, fairness, and privacy; transparency and explainability; robustness, security, and safety; and accountability. These principles support AI that is innovative but still dependable and respectful of people. They also emphasize that AI should benefit society as a whole, not just the organizations that build it.

10. How does the EU AI Act classify and regulate AI systems based on risk?

The EU AI Act uses a risk-based approach. AI systems considered an unacceptable risk are banned. High-risk systems — such as those used in employment, education, critical infrastructure, or essential services — must meet strict requirements for data quality, documentation, human oversight, testing, and monitoring. Systems with transparency risks must notify people when they are interacting with AI. Low-risk systems face fewer restrictions.

Section B: Applied and Scenario-Based

11. An AI recruitment tool consistently prefers male candidates. Which ethical principles are violated?

This violates fairness, nondiscrimination, accountability, and possibly transparency. If the tool favors male candidates, it likely learned biased patterns from past hiring data. Even if the bias wasn't intentional, the company is still responsible for finding out why it's happening. Applicants should also know when automated tools are being used and have a way to question or appeal the decision. Regular bias testing and human review would be needed before the tool should keep being used.

12. A self-driving car must choose between hitting one pedestrian or five. Which ethical dilemma does this represent?

This is the classic trolley problem — a moral dilemma about choosing between harming one person or allowing greater harm to several people. It raises questions about whether AI should focus only on minimizing total harm or whether other moral duties matter. In reality, self-driving cars should be designed to avoid dangerous situations altogether, not to make simple life-and-death trade-offs. But the scenario shows how hard it is to turn human moral judgment into software.

13. In predictive policing, how can bias be introduced into the algorithm's training data?

Bias often comes from historical arrest records. Arrest data doesn't show where all crime happens — it shows where police were already patrolling. If one neighborhood had more police presence in the past, the algorithm may label it as "high risk" and send even more patrols there. That creates a feedback loop. Underreporting, inaccurate records, and using variables like location or income as stand-ins for protected traits can make the bias worse.

14. How might a human-in-the-loop system reduce ethical risks in automated decision-making?

A human-in-the-loop system lets a trained person review, approve, or override an AI recommendation. This helps when the model misses important context or produces an unusual result. For example, an AI might mark a medical image as normal, but a doctor could notice symptoms the system didn't catch. Human involvement also makes accountability clearer. But the human reviewer needs enough time, training, and authority to disagree with the system when necessary.

15. Describe how data privacy laws such as GDPR affect the design of AI systems.

GDPR requires organizations to have a lawful reason for collecting personal data and to limit collection to what's actually needed. Developers must think about privacy from the start, secure the data, and explain how it's being used. People may have rights to access or correct their information and, in some cases, challenge decisions made only by automated systems. High-risk processing may also require a privacy impact assessment and meaningful human oversight.

16. A generative AI creates a fake video of a politician. What ethical issues arise here?

A fake political video raises issues around misinformation, consent, manipulation, and public trust. People might believe the politician said or did something that never happened. If it spreads during an election, it could influence voters before the truth comes out. There are also questions about who created it, who shared it, and who should be held responsible. Clear labeling, content authentication, platform rules, and penalties for harmful impersonation can help reduce the risk.

17. What are the ethical implications of AI-powered surveillance systems in public spaces?

AI surveillance can improve safety, but it also reduces privacy and can make people feel constantly watched. Facial-recognition errors may lead to innocent people being questioned. Some communities may be affected more than others. There's also the risk of "function creep," where technology meant for one purpose is later used for something more intrusive. Public agencies should show that surveillance is necessary, limited, accurate, and subject to independent oversight.

18. How can organizations ensure transparency in AI-driven financial decisions?

Organizations should clearly tell customers when AI is used in lending, insurance, fraud detection, or credit decisions. If someone is denied a loan, they should get a clear explanation of the main factors behind the decision. Companies should document the model, keep audit logs, test for discrimination, and allow customers to correct inaccurate information. There should also be a human appeal process. Saying "the algorithm decided" is not enough.

19. Discuss the ethical concerns surrounding AI in healthcare diagnostics.

AI can help detect illnesses earlier, but mistakes can directly affect a patient's health. A model trained on a narrow group of patients may not work well for others. Privacy is also a major concern because medical data is extremely sensitive. Doctors and patients should understand the model's limits, and a medical professional should always be involved in the final decision. From my EMS experience, real patients don't always match textbook examples. AI can support clinical judgment, but it shouldn't replace it.

20. What measures can be implemented to make AI datasets more representative and fair?

Organizations can collect data from a wide range of regions, age groups, racial and ethnic groups, income levels, genders, and people with disabilities when relevant. The data should be checked for missing groups, incorrect labels, and historical bias. Developers can use stratified sampling, rebalancing, bias testing, and external audits. Including people from affected communities in the design process also helps. Dataset limitations should be documented, and fairness should be monitored after deployment.

Section C: Reflective and Future-Oriented

21. Do you think AI can ever be completely unbiased? Justify your answer.

I don't think AI can ever be completely unbiased. These systems learn from human data, and our society already has gaps, inequalities, and disagreements about what "fair" even means. Developers also make choices about what data to use and what outcomes to optimize, and those choices can introduce bias even when no one intends it. So instead of expecting perfect neutrality, the realistic goal is to find bias, reduce it, measure its impact, and fix issues as they come up.

22. How can ethical frameworks ensure responsible innovation in AI?

Ethical frameworks give teams a checklist of things to think about before releasing a system. They push developers to look at privacy, safety, fairness, security, environmental impact, and how the system could be misused. They also encourage documentation, human oversight, and bias testing. This doesn't slow innovation — in many cases it actually improves the final product because problems get caught early, when they're easier to fix. Ethics should be seen as part of good engineering, not something that gets in the way.

23. Should governments enforce stricter AI regulations, or should industries self-regulate? Discuss.

I think the best approach is a mix of both. Companies understand their own technology and can create technical standards faster than governments. But relying only on self-regulation creates a conflict of interest, because companies may prioritize profit or speed over safety. Governments should set minimum rules for high-risk areas like healthcare, employment, finance, transportation, and law enforcement. Then industry groups can build more detailed standards that meet or go beyond those requirements.

24. What ethical challenges might Quantum AI pose in the future?

Quantum AI could solve certain problems much faster than today's systems, but that power comes with risks. It could break current encryption methods, putting financial, medical, military, and personal data at risk. Access might also be limited to powerful governments or large corporations, widening the gap between those who control advanced technology and everyone else. It could also make surveillance, cyberattacks, and automated decisions harder to detect or challenge. Ethical planning needs to start before the technology becomes widespread.

25. How might Generative AI reshape intellectual property rights?

Generative AI raises tough questions about training data, copyright, and ownership. Models may be trained on books, art, music, software, or photos without creators fully knowing how their work is being used. It's also unclear who owns an AI-generated result — the user, the developer, or no one. Copyright laws may need clearer rules about licensing, attribution, compensation, and how much human creativity is required for protection. The challenge is protecting creators without shutting down useful innovation.

26. What are the moral responsibilities of AI developers toward end-users?

Developers have a responsibility to avoid foreseeable harm and to be honest about what their system can and can't do. They should protect user data, test the system across different groups, fix known problems, and avoid releasing unsafe features just to beat a competitor. People should also have a way to question important automated decisions. Following orders doesn't remove personal responsibility — if a developer sees a serious risk, they should document it, raise the issue, and push for a fix.

27. How can AI sustainability contribute to global ethical goals?

AI sustainability means reducing the energy, water, hardware, and environmental resources needed to build and run AI systems. Developers can use smaller models when possible, improve data-center efficiency, reuse equipment, and track environmental costs. AI can also help the environment by improving power grids, monitoring pollution, predicting equipment failures, and supporting renewable energy. Ethical AI should consider both sides: how AI can help the planet and how its own resource use might harm it.

28. What future jobs might emerge to address ethical issues in AI systems?

We may see roles like AI ethics officers, algorithm auditors, AI governance specialists, fairness engineers, model risk managers, responsible-data curators, AI safety engineers, and digital-rights attorneys. Some people may specialize in investigating automated decisions or certifying that systems meet fairness and safety standards. Fields like healthcare, finance, robotics, and government may also need ethics experts who understand both the technology and the industry. These jobs will help connect software development with law, policy, and real human impact.

29. In your opinion, what will be the most pressing AI ethics issue in 2035, and why?

I think the biggest issue will be keeping humans accountable as autonomous AI systems take on more real-world responsibilities. By 2035, AI might be involved in transportation, healthcare, finance, communication, cybersecurity, and public infrastructure. When multiple models and organizations are involved, it becomes harder to figure out who is responsible when something goes wrong. We'll need clear limits on what AI can decide on its own, reliable audit trails, emergency shutdown options, and named humans who remain legally and morally responsible.

30. Reflect on your role as a future professional. How would you ensure ethical practices in AI development?

As someone going into AI and robotics, I'd treat ethics as part of the development process from the start. I'd test systems for safety and bias, protect private data, document limitations, and make sure important decisions can be reviewed by a human. My EMS experience taught me that real situations don't always match textbook examples, so technology has to work in messy, unpredictable environments. I'd listen to the people affected by the system and speak up when shortcuts create unnecessary risks. A system isn't successful if people can't trust it or if it causes preventable harm.